NOCTUO

PRODUCT CARD

PLA Silk

1. GENERAL INFORMATION ABOUT THE PRODUCT

PLA Silk is a PLA-based 3D printing material containing special color additives that give prints an elegant, satin sheen. The resulting surface is noticeably smoother, which effectively reduces the visibility of layer lines on model walls. The filament stands out with its attractive appearance and deep, vivid colors, making it ideal for projects focused on aesthetics. An additional advantage is trouble-free printing.

Main PLA Silk features:

- satin, elegant finish on the print surface,
- ready for use in almost any FDM 3D printer,
- good layer coverage,
- rich, vivid colors,
- parameters similar to classic PLA.

2. TECHNICAL PARAMETERS

CHARACTERISTICS	TEST METHOD	TEST CONDITIONS	IU	VALUE
PHYSICAL	**ASTM**			
Density	D792	-	g/cm^3	1.22
MECHANICAL	**ISO**			
Tensile strength	527	-	MPa	46
Breaking stress	527	-	MPa	59
Elongation at break	527	-	%	10
	ASTM			
Bending strength	D790	-	MPa	74
Notched impact strength, IZOD	D256	-	J/m	8
THERMAL				
Deflection temperature under load	E2092	0,45 MPa	°C	50

Tests have been done in 23°C, if it's not marked differently.

3. RECOMMENDATION OF PRINTING

PLA Silk does not require drying before using. Satisfying results could be achieved by using standard settings for PLA in available slicers and printers.

Cooling during printing is recommended.

Recommended parameters of printing:

Hotend temperature	220 - 230 °C
Bed temperature	50 - 60 °C
Print speed	< 300 mm/s

4. SAFETY NOTES

Exhaust fan is recommended.
Air filters in printer is recommended.
PLA needs to be used only in well ventilated conditions.
Inhaling fumes generated during the printing must be avoided.

Generating fumes during the printing depends mainly on printing temperature. In case of visibly raising emission level, the printing needs to end. Check the hotend temperature and efficiency of the control system before using it next time.

In proper using conditions, the product doesn't endanger health.

It's forbidden to set fire or exceed decomposition temperature!

Decomposition of PLA Silk is typically over 250 °C and it should be strictly avoided.
Main ingredient of decomposition is lactide.

Detailed safety information available in SDS.